



Examiners' Report Lead Examiner Feedback

Summer 2024

Pearson BTEC Nationals
In Computing (31769H)
Unit 2: Fundamentals of Computer Systems

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Grade Boundaries

What is a grade boundary?

A grade boundary is where we set the level of achievement required to obtain a certain grade for the externally assessed unit. We set grade boundaries for each grade, at Distinction, Merit and Pass.

Setting grade boundaries

When we set grade boundaries, we look at the performance of every learner who took the external assessment. When we can see the full picture of performance, our experts are then able to decide where best to place the grade boundaries – this means that they decide what the lowest possible mark is for a particular grade.

When our experts set the grade boundaries, they make sure that learners receive grades which reflect their ability. Awarding grade boundaries is conducted to ensure learners achieve the grade they deserve to achieve, irrespective of variation in the external assessment.

Variations in external assessments

Each external assessment we set asks different questions and may assess different parts of the unit content outlined in the specification. It would be unfair to learners if we set the same grade boundaries for each assessment, because then it would not take accessibility into account.

Grade boundaries for this, and all other papers, are on the website via this [link](#)

Introduction

Unit 2 (Fundamentals of Computer systems) for BTEC Level 3 National in Computing became available for first teaching in September 2016. This is the thirteenth time this assessment has been sat by learners. Examination opportunities will continue to be available for this unit twice a year in January and May/June. This unit is a mandatory unit for all learners studying either the Extended Certificate (360 GLH), Foundation Diploma (510 GLH) or Extended Diploma (1080 GLH).

This unit, along with Unit 1 (Principles of Computer Science), are assessed through a written examination paper. The examination is designed to test learners' understanding of computer systems within a range of contexts. The paper is divided into four main questions, each with a number of sub parts. Each main question is based around a unique scenario; each scenario is outlined at the beginning of that question and additional information and/or stimulus is provided with individual parts as required.

While appropriate credit is given for learners who demonstrate appropriate 'stand-alone' knowledge, more successful learners can apply their understanding to the scenarios provided in the question.

The paper is designed to assess the full grade range of the qualification. As such the paper is ramped so that it gradually increases in difficulty as the questions progress with a higher percentage of 'Pass' targeted marks in the earlier parts of the paper and the higher-grade questions towards the end.

Introduction to the Overall Performance of the Unit

While detailed analysis of specific questions in the paper appears later in this report it should be noted that overall learner performance was slightly weaker, with a drop in the mean mark compared to the most comparable series (June 2023). However it should be noted that this series had a significant percentage of entrants that were resit learners, which may have contributed to the drop in mean mark.

As has been mentioned in previous LE reports, the continues to be significant numbers of learners who were not sufficiently prepared for this examination and struggled to access marks. Understanding of the basic subject knowledge and vocabulary has been identified as an area of concern in previous examinations; it is clear that some centres have sought

to address this and as such these learners perform significantly better than their peers. However, there are still too many learners that lack understanding of basic and fundamental area of the subject content that is expected for a Level 3 Computing qualification.

Generally, learners demonstrated an understanding of the requirements of different command verbs, and in Explain and Describe questions for example, would generally construct questions in an appropriate manner. However, as previously in many cases a lack of technical understanding, and in some case a lack of depth in knowledge, prevented many learners from accessing marks.

In higher tariff questions many learners were not able to fully connect points or provide expansions in a meaningful way. While it is clear that many learners have made progress in this area, where an Medium open response carries a tariff of 3 or 4 marks, it is very rare that learners will provide a response that goes beyond 2 marks. Centres are encouraged to work on this area with learners, in particular focusing on building on a response in a way shows deeper understanding of the point made by ensuring each connected point made builds on the previous point.

Learners continue to perform less well in extended open response questions (e.g. Discuss, Analyse, Evaluate). As with the medium open response questions, responses often lacked depth of knowledge and where knowledge was sound responses were rarely applied to the given scenario is also often lacking.

Sample assessment materials, previous papers and sample marked learner work with learners, are available on the Pearson website and centres should make use of these to help prepare learners for the demands of the examination. Accessing and exploring these materials will help ensure that learners are familiar with the design and expectation of the paper. However, these should be used with a small note of caution; centres should avoid using these to teach purely rote answers, but instead should use these to explore examination technique, in particular how to decode the meaning of questions and apply knowledge that has been taught in the classroom within the given context of an examination question.

Individual Questions

Question 1a

Overall, this question was very poorly answered, with a large proportion of learners not accessing any marks. This particular question item was designed to be a quite accessible recall question, requiring learners to identify advantages of using a desktop PC compared to using a laptop. As with a number of questions in this paper, it was clear however that large pockets of learners had limited understanding of core knowledge and vocabulary in the unit. Many learners provided generic responses such as "faster", "more storage", "more memory", which did not gain credit. The specifications of any computer varies from device to device, for example a high-end laptop would out perform a budget desktop in terms of speed and storage. Learners that performed well on this question focused on the impact of the form factor of the devices.

Q1a Example response 1

(a) Ruben has provided desktop PCs for use by the staff and children.

Give **three** advantages of desktop PCs compared to laptops.

(3)

1. The parts that desktop pc's are made out of are a lot more durable compared to laptops.
2. There is little to no risk of theft for pc's, however laptops are portable so anybody can pick them up and walk away.
3. Desktop pc's use much more powerful parts in comparison to laptops, resulting in a lot more data being able to be processed at a much faster rate.

Lead examiner commentary:

1. No awardable content – depending on the device/situation laptops or desktops could be more durable.
2. "Little risk of theft" (1) – mark point 6.
3. Not enough to award mark point 2 – Must identify that you can get a similar or more powerful desktop cheaper than a comparable laptop.

1 mark awarded

Q1a Example response 2

(a) Ruben has provided desktop PCs for use by the staff and children.

Give **three** advantages of desktop PCs compared to laptops.

(3)

- 1 DESKTOP PC has bigger monitors for viewing purposes ~~where~~
- 2 It is a cheaper option as it allows multiple users in the system instead of buying a laptop for each one
- 3 ~~Desktop~~ desktop PC are more easily repaired compared to laptops as they are more spare parts that can be replaced in a PC compared to laptops.

Lead examiner commentary:

1. Mark Awarded – Mark point 5.
2. No awardable content – reference to cost/cheaper must be clarified. I.e. cheaper for the **same (or higher)** specification
3. Mark awarded - "more easily repaired".

2 marks awarded

Question 1b

Performance on this question was sporadic, with many learners not effectively accessing marks. A lack of technical vocabulary in many cases prevented learners from accessing marks as they were not able to clearly express their meaning. Where learners did access marks, response tended to focus on formatting techniques, rather than **features** of word processing software, so although some credit was given for example of formatting techniques this did limit the spread of marks learners achieved.

Question 1c

Performance on this question was overall quite poor. As mentioned in other places in this report, many learners lacked meaningful understanding of the content being assessed; in this case the nature of open-source software.

Many responses focused on security concerns, but a very common misconception was that anybody could access the active, installed version being used, rather than the availability/visibility of the original source code from the developers, which may allow bad actors to develop exploits.

Q1c Example response 1

(c) Ruben has chosen to use open source word processing software in the nursery.

Explain **two** possible drawbacks of using open source software to create the records.

(4)

1 Open source software means anyone can have access to it which can raise threats of it getting hacked

2 Another drawback can be that anyone can make changes to which can result in inaccuracies as someone might have changed the information.

Lead examiner commentary:

Response 1

"...anyone can have access to it" – Not enough to award mark – must show understanding of access to original code base.

"Which can raise the threat if getting hacked" - (1 mark) Just enough to award identification mark MPT 5.

Response 2

No awardable content – The level of access/ability to change information not relevant to the question context i.e. open-source code.

1 mark awarded

Q1c Example response 2

(c) Ruben has chosen to use open source word processing software in the nursery.

Explain **two** possible drawbacks of using open source software to create the records.

(4)

1. There isn't as much support with open as there is proprietary. So if Ruben has any issues he has to fix it ~~for~~ himself or hope.

2. open source can be tricky to navigate without training so Ruben will have to spend extra time training his staff on how to use the system.

Lead examiner commentary:

Response 1

"Isn't much support" (1) Just enough to award identification mark for MKPT 4.

Response 2

"May be tricky to navigate" (1) "have to spend extra time training staff" (1 mark)

Alternative wording for MKPT 2 identification and expansion.

3 marks awarded

Question 1d

Performance on this question was overall very poor. Many learners were clearly unfamiliar with the use of matrices and as such were unable to demonstrate their use within a calculation. Many learner responses were left blank, or provided a simple numerical calculation which did not utilise matrices. Where learners did access marks, the final total was often represented as a matrix, but the process was omitted, limiting learners to 1 mark out of 2.

Question 1e

Performance on this item was very polarised, where many learners lacked an understanding of core terminology; in this case "row-major order". As such those that understood the term were able to respond correctly to the question. A note for further development, the item also required the data to be re-written as an array, and although learners were not penalised in this instance, many merely re-wrote the data as a list and did not demonstrate it as an array (i.e. Within brackets, separated by commas).

Question 1f

Performance on this question was quite poor, with many learners lacking in-depth knowledge of the content. In this case arrays, specifically two-dimensional arrays and how they hold data. Centres are encouraged to explore this area of the specification in more depth both from a general standpoint (e.g. What a 2D array is, how it works, comparison to 1D arrays) and then expand on how arrays are used in various contexts.

Q1f Example response

(f) **Figure 3** shows the data for three other months represented as a two-dimensional array.

January	135	161	158	102	96	103
February	108	140	153	138	115	110
March	121	138	120	109	125	123

Figure 3

Explain **two** reasons why a two-dimensional array is a suitable data structure to store this data.

(4)

1. Because each Row will have its own personal number so that when a user wants to find the data they need, they can input the number the data belongs too.
2. the data is easily stored because of how simple and effective this data structure is.

Lead examiner commentary:

Response 1

"Each row will have its own number" (1) – Just enough to award against "index will always be the same" (MKPT 3).

"To find the data input the number the data belongs too" [sic] (1) – awarded against MKPT 3 expansion.

Response 2

No awardable content.

2 marks awarded

Question 1g

Performance on this question was generally fine. This question was designed to be the most difficult item in this section, so it would be expected that some learners would not access as many marks. However, many learners were able to access at least one of the three marks available, responses tended to focus on preventing errors, but then often lacked deeper understanding. A common misconception displayed here was that validation ensures data is correct/accurate rather than ensuring that it is sensible. Many learners provided examples of validation checks, which while they did gain some credit by demonstrating an understanding of ensuring data follows rules, descriptions of checks did not meet the requirements of the question, i.e. **why** data is validated and not **how** it is validated.

Q1g Example response 1

(g) The attendance data goes through data validation functions before it is processed.

Explain why data should be validated.

(3)

It is validated to check if the data is in the right format and range. An error will pop up if one of these is wrong which helps to not make mistakes.

Lead examiner commentary:

“Check data is in the right format” (1) – if data does not meet requirements.

“An error will pop up” (1) – by alerting the user.

“Helps to not make mistakes” – not enough to award “avoid errors” validation does not prevent mistakes/incorrect data.

2 marks awarded against MKPT 2.

Q1g Example response 2

(g) The attendance data goes through data validation functions before it is processed.

Explain why data should be validated.

(3)

Data must be validated because it will keep the data accurate ensuring the data is correct.

Lead examiner commentary:

No mark awarded

Validation ensures data is sensible, it does not guarantee that it is accurate/correct.

Question 2a

A number of learners misinterpreted this question and as such did achieve marks here. The question required the learners to identify how a device could be connected to a cloud server when the workers are away from the office. Many provided response that just focused on general ways of connecting to cloud services (e.g. Using the internet, using an app) rather than how the connection could be made when working on the move. Where learners did focus on a connection, many did not move beyond "Wi-Fi" and as such gained 1 out of a possible 2 marks.

Q2a Example response 1

(a) State **two** ways that the nurses could connect their devices to the cloud-based server when they are not in the office.

(2)

1. By through mobile connection and internet connection
2. They could use a drive USB that transfers files to the server

Lead examiner commentary:

1. "Mobile connection" not enough – should be using correct terminology for connection types.
2. No awardable content.

0 marks total

Q2a Example response 2:

- (a) State **two** ways that the nurses could connect their devices to the cloud-based server when they are not in the office.

(2)

- 1 it could be connected through remote access using the patient's internet connection in their homes
- 2 if the nurses have mobile broadband they can use the device and connect to the server from the office

Lead examiner commentary:

1. "Patients internet connection in their homes" (1) – MKPT 3 – just enough to show an understanding of using a wi-fi or temporary network connection.
2. "Mobile broadband (1)

2 marks awarded

Question 2b

Performance on this question was quite sporadic. Many learners focused (incorrectly) on accessing data remotely. The question instead required the learners to think in context as to why *this* charity would choose a cloud-based server, so for example that they would not have the technical expertise required to maintain a physical server, without employing additional staff.

Q2b Example response 1:

- (b) The cloud-based server allows staff to access data held on the server using any device if they have an internet connection.

Explain **two other** reasons why the charity would choose to use a cloud-based server.

(4)

- 1 One other reason is that its more secure platform and there data cannot be corrupted.
- 2 Another reason is that it provides a large amount of storage.

Lead examiner commentary:

Response 1

"More secure" (1) just enough to award identification MKPT 5.

Response 2

No awardable content – reference to storage/resources must show an understanding of scalability.

1 mark awarded

Q2b Example response 2:

- (b) The cloud-based server allows staff to access data held on the server using any device if they have an internet connection.

Explain **two other** reasons why the charity would choose to use a cloud-based server.

(4)

1 it is cheaper to use a cloud based server as you can buy as many servers as you want because of scalability and you don't have to buy all of the server hardware.

2 you would have more physical space as physical servers are quite big and take up a lot of space.

Lead examiner commentary:

Response 1

"Cheaper..." (1) "...don't have to buy the server hardware" (1 mark) Enough to award both marks against MKPT 6 as it is clear they are talking about the initial costs.

"As many servers as you want" – no mark. Responses must show an understanding of responding to need.

"Scalability" (1 mark)

Response 2

"Have more physical space" - a valid point (MKPT 6). However, this mark has already been awarded.

3 marks awarded

General marking guidance

Two responses can be awarded from the same answer space, but learners are encouraged to ensure that their responses are more clearly presented to ensure their answers are clear to examiners.

Question 2c

Question performance was overall quite good. The majority of learners were able to successfully name two suitable utility software applications and many of those were able to provide suitable expansions to explain the way in which they protect devices and data. Where learners struggled to access marks, these responses were typified by a lack of understanding of technical vocabulary, specificity not being aware of what a “utility software application” and providing actions or processes to protect data instead.

Q2c Example response 1:

(c) In addition to the cloud-based server the charity requires a number of different physical devices (e.g. laptops, tablets).

Explain **two** utility software applications that could be used to protect the devices and data.

(4)

- 1 ~~Passwords that~~ Having accounts only accessible by passwords
- 2 2FA where the user has to verify it is them by ~~gan~~ entering a code from a trusted device or contact.

Lead examiner commentary:

No Awardable content.

Answers must specifically refer to utility software and not just general security approaches.

0 marks awarded

Q2c Example response 2:

(c) In addition to the cloud-based server the charity requires a number of different physical devices (e.g. laptops, tablets).

Explain **two** utility software applications that could be used to protect the devices and data.

1. Antivirus can protect the computer which has access to this information preventing any data leaks from the use of malicious techniques for example key logging. (4)

2. Encryption protects user information when received or sent from physical devices and can only be deciphered in a feasible amount of time using a decryption key.

Lead examiner commentary:

Response 1

Antivirus (1)

"Protect...malicious techniques for example key logging" (1)

Response 2

Encryption (1)

Expansion shows enough understanding of protecting transmitted data to award mark.

4 marks awarded

Question 2d

Overall, this question was answered very well with the majority of learners gaining two out of the three available marks. Typically, learners were able to provide a suitable linked description of how compression works e.g. "Removes data (1) which reduces file size (1)", very few learners however were able to provide a third linked step. Centres are encouraged to build on good performance in such questions and explore how learners can respond and build upon step/point made.

Q2d Example response 1:

(d) Data compression is used when transferring data between the cloud-based server and connected devices.

Describe how data compression could improve data transmission speed.

(3)

Data compression decreases the size of file by removing repeated files ~~data~~ or not needed data. The smaller size of file is, less time it takes to send ~~data~~. There are lossy and lossless data compression, ~~lossless~~ compression will ~~take~~ make the file smaller and ~~at~~ send it quicker.

Lead examiner commentary example 1:

"Removing repeated data" (1)

"Smaller size of file" (1)

The rest of the response is mostly a repeated point.

2 marks awarded

Q2d Example response 2:

- (d) Data compression is used when transferring data between the cloud-based server and connected devices.

Describe how data compression could improve data transmission speed.

(3)

Data compression causes a file to lose pixel
this then reduce the amount of storage
the file takes up making transmission
speeds faster as less data is being
transferred.

Lead examiner commentary example 2:

"Lose Pixels" (1) – removes data.

"Reduce the amount of storage file takes up" (1) reduced file size

"Less data is being transferred (1)

3 marks awarded

Question marking guidance

Credit responses that use alternative wording but show an understanding of the concepts covered in the mark scheme.

Question 2e

Performance on this question was very good with the majority of learners correctly identifying all three correct logical outcomes.

Question 2f

Performance on this question was in line with expectations. Typically, most learners were able to gain two or three marks out of a possible four. Where learners typically did not achieve full marks, this was through providing incorrect outcomes for the final test. However, the difficulty of this was designed to ramp from the previous tests in this question so as such it performed as expected. In addition to the overall good performance seen on this question, there are still unfortunately quite large pockets of learners who do not appropriately access flowchart questions. Centres are encouraged to work with learners on both interpreting given flow charts and producing flow charts to demonstrate processes and solutions.

Question 3a

Overall performance on this question was quite good, with the majority of learners able to access marks. Typically, learners gained one or two marks out of a possible four by showing an understanding of loss of picture quality/clarity and the reduction in overall file size. Where learners tended to do less well is with the more technical parts of the required answer, specifically that fewer bits are used per pixel which means fewer colours can be created.

Q3a Example response 1:

(a) Fran uses many images in her games.

Describe how reducing the bit depth of an image will impact on the way the image data is stored and displayed.

(4)

The image will take up more space when is being stored as ~~it~~^{she} is modifying it. The image might not be good quality as she has reduced the bit depth.

Lead examiner commentary:

"Will not be as good quality" (1) – Enough for MKPT 3.

1 mark total

Q3a Example response 2:

(a) Fran uses many images in her games.

Describe how reducing the bit depth of an image will impact on the way the image data is stored and displayed.

(4)

-reducing bit depth reduces file size
↳ so file takes up smaller storage capacity
-reducing bit depth reduces image quality
↳ as it has less pixels to store colours
↳ however, it may not be noticeable depending how much the bit depth has been reduced.

Lead examiner commentary example 2:

"Reduces file size" (1) – MKPT 4

"So takes up smaller storage capacity" (1) – MKPT 5

"Reduces image quality" (1) – MKPT 3

"Has less pixels to store colours" (1) – MKPT 1.

4 marks total

Question 3b

Learner performance on this question was quite limited. While many learners were able to gain some marks, responses rarely provided more than a definition of emulation such were unable to move beyond mark level 1 (2 marks). Where learners did explore the use of emulation in more detail, they often did not sufficiently link their response to the given context (i.e. How the use of emulation would impact on/be utilised by a games designer)

Performance on this question highlighted response characteristics which were present in many of the extended responses, and highlights some of the continued areas for development for learners and centres. As mentioned in previous Lead Examiner reports centres should consider these points:

1. The extended questions are an opportunity to demonstrate deeper knowledge. Centres should work with learners to develop ideas and expand on points made. Using examples and reasons where appropriate
2. More successful learners make use of the context provided. To access the higher mark bands knowledge from the specifications core concepts should be considered and applied to the give scenario.

Q3b Example response 1:

(b) Fran uses emulation when designing games.

Discuss the factors Fran will need to consider when using emulation.

You should support your discussion with examples that are relevant to the scenario.

(6)

Emulation is where you pretend that a device you are on is on a different platform. An example of this is if you wanted to play playstation games but are on a laptop, you could use emulation to make the game think you were on a playstation as it would copy it in every way. This process of emulation is slow and takes up

large amounts of storage. When designing a game cross platform so that it would be available on different consoles and devices would give a bigger audience although would cost more to design a game like this. Although it would add to the overall user friendliness.

Lead examiner commentary:

Trait 1 - Knowledge and understanding

Level 2(just) – There is a clear understanding of what emulation is, but limited understanding of related factors.

Trait 2 - Relevance to the context

Level 1(just) –Although designing a game is mentioned, it discusses general design principles rather than how **emulation** is used in the process.

Trait 3 - Discussion

Level 1 – there is an attempt to build upon points made but they are generic and not related to the context of the question.

Using “best fit” (See marking guidance at start of MS) the response is placed at the top of mark level 1.

2 marks awarded

Q3b Example response 2:

(b) Fran uses emulation when designing games.

Discuss the factors Fran will need to consider when using emulation.

You should support your discussion with examples that are relevant to the scenario.

(6)

In simple terms emulation is ~~described~~ defined as installing a system inside another system. It is when you install a copy of another systems OS onto your computer and run it through a software known as an emulator.

Being a game designer emulation will prove useful to ~~test~~ Fran because it will enable him be able to see how his game works in another OS / different

systems, ~~the~~ thereby helping him make accurate decisions. without having to spend money to purchase the system and ~~the~~ run the game there thereby saving costs

Despite the numerous advantages it is important to note that emulation is not always legal and you could end up in trouble for it. Another thing to note is that is that emulated files don't always work well since they don't typically belong to your machines OS, they tend to crash and have issues. In severe cases they end to ~~cause trouble~~ destroy the system in which they were installed on thereby ~~causing~~ causing issues.

so before emulation Fran has to make sure the system/software he is emulating, is legal/allowed by the ~~pro~~ producer/manufacture and the file is safe/secure and his system can run it.

Lead examiner commentary example 2:

Trait 1 - Knowledge and understanding

Level 3 – The knowledge is accurate – a wide range of valid points made.

Trait 2 - Relevance to the context

Level 3 –All the points made would be relevant considerations when designing a game and using emulation. The learners attempt to link many of them back to the scenario.

Trait 3 - Discussion

Level 3 – all points are expanded and there are logical links.

The response fully meets the descriptors for mark level 3.

6 marks awarded

Question 3c

Performance on this question was on the whole very poor. Many learners failed to achieve any marks on this question, and of those that did few moved beyond providing answers on the level 1 and level 2 borderline (3-4 marks).

Learners typically lacked the technical understanding of the components of the operating system and were often unable to even provide fully correct definitions of what each of the functions are. Those that were able to demonstrate understand of the components, rarely applied this to the scenario (i.e. Executing a computer game).

Q3c Example response 1:

(c) Fran is developing a multiplayer online game.

Analyse how the operating system will manage system components and tasks when executing the game.

You may wish to consider (but are not limited to):

- networking
- multi-tasking
- interrupts
- memory management.

You should support your analysis with examples that are relevant to the scenario.

(8)

The operating system will connect to the internet when executing the game, this is because it will connect to a server so the user can interact with other users on the game. The operating system will multitask when executing the game, this is because it will have to detect your inputs when you are on the game, load the graphics and shaders and textures and show other players that are in the game with you.

It may need to manage memory to check that the latest version of this game is installed and if it isn't it may produce an error as everyone's versions need to be synchronised because it is an online game.

Lead examiner commentary example 1:

Trait 1 - Knowledge and understanding

Level 1 – Knowledge is very basic.

Trait 2 - Relevance to the context

Level 1 – Although there is an attempt to apply points made to the scenario, the lack of depth of knowledge means these examples are often weak or incorrect.

Trait 3 - Analysis

Level 1 – the points are broken down to some degree, but they are very generic. The response fully meets the level 1 descriptors, which describes the top of the band.

3 marks awarded

Q3c Example response 2:

memory management.

You should support your analysis with examples that are relevant to the scenario.

(8)

The operating system (OS) is likely going to have to treat the game as a real-time, full duplex communication scenario. The OS would need to level the priorities of the interrupts (e.g. inputs from a mouse or keyboard, and the type of input devices being used (perhaps a controller)). This will be controlled based on the control unit. The OS will put data from the game into a separate and organised folder in order to keep it easy to access for the user while avoiding accidental user deletion of key components of the game. The OS will need to have a large allocated storage for the game especially whilst it is being played. It will have to defrag storage in order for faster loading of the game and speedier reactions as the other components are less strained. Multi-tasking will be essential for the OS to control as there is a constantly changing input-output system to the games environment. Other apps

should be given less processor time and priority in order for the game to run smoothly. On a networking basis, the voice-over internet protocol must be followed (as well as e.g. TCP/IP) to work efficiently. Data packets from other players also need to be received while also allowing for communication.

(Total for Question 3 = 18 marks)

The OS will need to keep up with the online game server for any changes as well as doing logging the right content. Also, if possible the firewall should be controlled efficiently to allow lots of incoming packets but also prohibiting any viruses from other users.

Lead examiner commentary:

Trait 1 - Knowledge and understanding

Level 3 – The response shows a accurate and detailed knowledge of the way that the OS manages tasks.

Trait 2 - Relevance to the context

Level 3 – Although not all points are directly linked to the scenario, there is sufficient links shown to meet the level 3 descriptor.

Trait 3 - Analysis

Level 3 – the points are broken down effectively and points made are linked to the OS, the game or both.

The response fully meets the level 3 descriptor.

8 marks awarded

Question 4a

Performance in this question was generally quite well spread with a full range of marks accessed, although typically responses demonstrated the characteristics of mark level 1 / level 2 borderline (3-5 marks). While many learners were able to also achieve higher in level 2, significantly fewer learners were able to access mark level 3.

Where learners typically were able to access higher marks, this was due to a lack of application to the given scenario. While many learners were able to provide suitable ways hardware can be used to protect data, there were often little consideration of how this would apply in this context, e.g. how would implementing biometrics impact on staff and work processes in the context of a manufacturing company, or who should have access to the data & what might be the best way to ensure only those staff can access it.

In order to improve general performance with the extended open response questions, centres should work on examination technique and developing learners' breadth and depth of knowledge, so that it can be applied appropriately in a range of scenarios.

Q4a Example response 1:

- (a) Discuss ways that the company could use computer hardware to protect its data and computer systems.

You should support your discussion with examples that are relevant to the scenario.

(10)

The company could use biometric identification devices such as fingerprint and face recognition to protect the personal information and trade-secrets kept on its computer systems from unauthorised access, as they have a financial and legal incentive to protect it.

In addition to this, the company could use magnetic tape, or spare hard drives, or optical media to store copies/backups of their data so that in the event that it is damaged, stolen, or lost, they do not lose everything and are still able to operate.

Furthermore, the company may want to invest in more, off-site servers so that in the event that one server is damaged or out of service data can still be processed and accessed allowing them to continue operations.

Lead examiner commentary:

Trait 1 - Knowledge and understanding

Level 1 – There is some basic knowledge displayed about the use of computer hardware. E.g. biometrics. Reference to back up is only partially relevant. Back up is a process rather than how the **hardware** is used.

Trait 2 - Relevance to the context

Level 0 – The content is generic, no reference made to the requirements of the given company/scenario.

Trait 3 - Discussion

Level 1 – There are some attempts to expand upon the points made, but there are often generic and superficial.

The response partially meets the level 1 descriptor.

2 marks awarded

Q4a Example response 2:

- (a) Discuss ways that the company could use computer hardware to protect its data and computer systems.

You should support your discussion with examples that are relevant to the scenario.

(10)

The company could use computer hardware to protect ~~that~~ its data by locking unauthorised users out of the device by using a finger print recognition system. This means that ~~only~~, because finger prints are unique, only authorised personnel can get into the devices.

It's important to make sure that only authorised users can access data or it could easily get stolen, altered or deleted. This is why many companies have strict regulations in place to protect their vital data.

For example, ~~laptops~~ laptops have got a finger print

Scanner on them to keep unauthorised users out of the laptop.

Also another method of using hardware to protect data is by having the newest model in your devices because they are usually more efficient and will have ~~the~~ better security built in to them.

E.g new hard drives & SSD's will have less of a chance of crashing or getting errors which means accidental data loss is less likely.

Lead examiner commentary:

Trait 1 - Knowledge and understanding

Level 2 – The response covers suitable hardware components (Fingerprint scanner) and reconsider impact on data.

Trait 2 - Relevance to the context

Level 1 – There is an attempt to link the hardware concepts (specific device, and hardware reliability) to the idea of data security, but there is little consideration within the given example company.

Trait 3 - Discussion

Level 2 (Just) – The points made are expanded well in places however these are quite generic.

The response partially meets the level 2 descriptor.

5 marks awarded

Question 4b

As with Q4a performance on this question was quite mixed. Whilst many learners were able to provide some generic benefits and drawbacks of the given backup and storage systems, responses were often quite generic and superficial, which led to basic and generally not well supported conclusions. As such, most learners' responses were limited to limited to the top of mark level 1, or the bottom of mark level 2.

More successful learners were able provide a more nuanced evaluation, and considered the suitability of the backup systems in the context of the given company and the data that they hold.

Q4b Example response 1:

Evaluate the suitability of the company's current backup and storage systems.

You should support your evaluation with examples that are relevant to the scenario.

~~The~~ The companies use of ^{storage devices (12)} ~~active data~~ for ~~storage~~ active data could be improved as well as ~~the~~ their backup schedule.

The companies use of SSD's for ~~eg~~ ~~active data~~ ~~sto~~ ~~storing~~ storing active data is okay as long as the storage capacity is high. SSD's allow for fast access to the data due to fast read and write speeds but they are expensive when compared to HDD's. They are more reliable than HDD's due to no moving parts but HDD's allow for higher storage capacity. The SSD will be best used for frequently opened data.

The use of HDD's by the company could be improved by mainly using them for

larger, not regularly opened files due to there slow read write speeds when compared to SSD's. HDD's also have high capacity and are a lot cheaper than SSD's so the company could have ~~on~~ more storage space.

The companies current backup system needs improvement. ~~For~~ To start, saved data is only being backed up once a week so data that has been changed and added during the time period to the next backup could end up being lost if the ~~system~~ ^{System} ~~crash~~ ^{crash} or end up damaged. It would be better to introduce a differential and incremental backup into the companies system as this will ensure that data that has been changed will be backed up before the next full backup.

(Total for Question 4 = 22 marks)

TOTAL FOR PAPER = 80 MARKS

Finally to improve the backup system it would be better for the company to either use the cloud to store there backups or have another server in a different building just in case of a fire or flood therefore they will still have there data so they could recover a lot quicker and easier

Lead examiner commentary:**Trait 1 – Vocabulary and arguments**

Level 3 –clear and accurate use of appropriate technical vocabulary used to support accurate knowledge.

Trait 2 - Relevance to the context

Level 2 (just) –The response considers both storage considerations and makes reference to the procedures (active and back up) but there are occasional misunderstandings (back up frequency) and only occasional consideration of the company's needs.

Trait 3- Evaluation

Level 2 –Judgments are made throughput and alternative solutions are suggested. Although there is no distinct final conclusion, the final paragraph does support the partially developed points made though out the response.

Using “best fit” the response is placed at the top of **mark level 2**.

8 marks awarded

Q4b Example response 2:

Evaluate the suitability of the company's current backup and storage systems.

You should support your evaluation with examples that are relevant to the scenario.

(12)

SSD is a new storage device that stores data using flash based memory which is much faster than any traditional hard disk. The flash ~~means~~ memory inside an SSD means data is written, transferred and erased electronically and also silently, which is very good when it comes to managing data and storage because it all saves and if data is somehow lost, the written data is saved. External storage is good to backup data because data outside of a computer or other digital devices are saved it also provides additional space for storing files, documents, media and other types of data which is good for the company.

Lead examiner commentary example 2:

Trait 1 – Vocabulary and arguments

Level 1 – Some Accurate technical vocabulary used, but many of the points made are vague. Some basic consideration of what the devices do.

Trait 2 - Relevance to the context

Level 1 (just) – Very limited consideration of the devices in relation to the backup procedures of the company, beyond a very superficial link to storing data.

Trait 3- Evaluation

Level 1 (just) –some basic VERY basic value judgements have been made.

The response partially meets the descriptor for band 1.

2 marks awarded

Summary

The learners' overall performance for this unit during this summer was slightly poorer than the comparable Summer 2023 series in terms of mean mark.

There were significant instances, where learners achieved 0 marks, due to significant gaps in terms of core knowledge and understanding. This resulted in blank responses or answers that just did not address the requirements of the question.

Performance across the paper suggested there are many learners that lack the basic technical understanding expected in a level 3 Computing qualification. However, there is some evidence of very good learner performance, and centres should continue to build on this to ensure learners have both the breadth and depth of knowledge required for success in this examination.

Centres are encouraged to consult the 'Technology Update' which will be published on the BTEC website. This document defines the scope of the technologies that may be used in examinations such as defining the range of 'common protocols', 'Input devices' 'utility software' etc. and should be used in conjunction with the specification when planning and delivering content.

Based on performance in this examination series, learners are offered the following advice to help improve performance:

- Develop a stronger understanding of key terminology used in the unit so that you are able to access the context of the question.
- Explore some of the more technical aspects of the specification in greater depth.
- Ensure that when providing answers/information your response is applied to the given context.
- Continue to improve understating of the requirements of the different command verbs used in the unit, particularly the extended questions, so that you can structure your response appropriately and maximise the marks you achieve.
- Further support on the requirements of command verbs can be found in the specification and in training materials published on the Pearson website.
- For shorter response questions (5 marks or less), make note of the number of marks available this will help you identify the number of points you need to make. For example, a 4 mark 'Explain one...' style question would need to make at least four linked points, three of which expand/exemplify understating of a single point.
- Develop the quality of responses to extended writing responses (6 marks or more). Ensure you consider a range of points, each of which should be expanded or supported with examples and applied to the given context. General/generic answers often struggle to meet descriptors beyond that of mark band 1



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